# Optimized Moirai - Paper Notes and Roadmap

Reference Paper Notes and Research Roadmap

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# Part 1 — Study Notes on the Three Reference Papers

## Paper 1: Moirai 2.0

**Moirai 2.0: When Less Is More for Time Series Forecasting** Chenghao Liu, Taha Aksu, Juncheng Liu, Xu Liu, Hanshu Yan, Quang Pham, Silvio Savarese, Doyen Sahoo, Caiming Xiong, Junnan Li — Salesforce AI Research, arXiv:2511.11698v3, Feb 2026, 16 pages.

**Core idea:** Simplify the Moirai architecture rather than scale it up.

**Architectural changes from Moirai 1.0 to 2.0**

| Component | Moirai 1.0 | Moirai 2.0 |
|---|---|---|
| Backbone | Encoder-only, masked reconstruction | Decoder-only, causal/autoregressive |
| Patching | Multiple patch sizes | Single patch size |
| Output head | Mixture-of-distributions | Direct quantile (pinball loss) regression |
| Token prediction | Single-token | Multi-token (predicts several future patches per step) |

**Training recipe (useful for your Methods section):**

- Pretraining corpus: 36M time series, ~295B observations
- 100,000 training steps, batch size 256
- Optimizer: AdamW, lr = 1e-3, weight decay = 0.1, beta1 = 0.9, beta2 = 0.98
- Linear warmup over 10,000 steps, then cosine annealing
- bf16 mixed precision
- 50% of input patches randomly masked during training for robustness
- Normalization statistics computed from only the **first 30%** of each sample, to avoid leaking future information into the causal decoder — a leakage-avoidance trick worth adapting for your own pipeline

**Loss function:** quantile (pinball) loss averaged over 9 quantile levels (0.1 to 0.9), directly aligned with the CRPS evaluation metric.

**Results:** Ranks 5th–6th of 37 foundation models on the GIFT-Eval benchmark (MASE/CRPS). 2x faster and 30x smaller than Moirai 1.0-Large while performing better. Scaling up further does **not** help — larger variants underperform the small model, and accuracy degrades at longer horizons. The authors flag this openly as unresolved.

**Why it matters for your project:** This is the current state-of-the-art lineage of the base model you are extending. Your `OptimizedMoiraiModel` is architecturally closer to Moirai 1.0 (encoder-based, extra heads added on top) than to 2.0's decoder-only design. Your paper needs one explicit paragraph justifying why you build on 1.0's encoder rather than 2.0's decoder-only architecture.

## Paper 2: Moirai-MoE

**Moirai-MoE: Empowering Time Series Foundation Models with Sparse Mixture of Experts** Xu Liu, Juncheng Liu, Gerald Woo, Taha Aksu, Yuxuan Liang, Roger Zimmermann, Chenghao Liu, Silvio Savarese, Caiming Xiong, Doyen Sahoo — Salesforce AI Research / NUS / HKUST, arXiv:2410.10469v1, Oct 2024.

**Core idea:** Moirai 1.0 specializes by sampling **frequency** — a separate input/output projection layer per frequency (hourly, daily, etc.). The authors argue frequency is an unreliable proxy for pattern, because:

1. Series at different frequencies can look alike.
2. Series at the same frequency can look completely different.
3. A single series is non-stationary — its own distribution shifts within one context window.

**Fix proposed:** one shared input/output projection layer for all data, with a sparse **Mixture-of-Experts (MoE)** layer inside the transformer so specialization happens per-**token**, learned automatically from data rather than assigned by a human-picked frequency label. They also introduce an improved gating function based on cluster centroids from a pretrained model, rather than a randomly-initialized gate.

**Results:** Up to 17% improvement over Moirai 1.0 at equal model size, across 39 datasets (in-distribution and zero-shot), using up to 65x fewer *activated* parameters than competing foundation models.

**Why it matters for your project:** This is the closest methodological relative to your own contribution. Your `use_regime` flag (an intraday "session regime" embedding) is a lightweight analogue of the same insight — instead of full MoE routing, you inject a hand-crafted regime signal.

> ***Suggested framing for your paper:*** *"Moirai-MoE lets the model discover specialization automatically via sparse routing; we instead inject a domain-informed regime embedding, trading some flexibility for interpretability and much lower compute cost, which is appropriate for a single-market, single-frequency intraday setting."*

## Paper 3: Multimodal Financial Forecasting Survey

**Learning Across Modalities: A Systematic Survey of Multimodal Models for Financial Analysis** Simone D'Amico, Fabio Mercorio, Navid Nobani, Giancarlo Sperli, Carmine Ventre — University of Naples Federico II / University of Milan-Bicocca / King's College London. In press, *Information Fusion* (Elsevier).

**Not a model paper — a systematic map of the field.** Reviews 35 papers (2018–2025) along four axes:

1. **Input modality** — price/volume time series, text/news, relational/graph data
2. **Architecture family** — RNN/LSTM/GRU, GNN, hybrid models
3. **Fusion strategy** — early fusion (raw inputs combined), intermediate fusion (learned representations combined), late fusion (model outputs combined)
4. **Predictive task** — price prediction, return prediction, stock ranking

**Field-wide challenges identified** (directly usable for your own "why is this hard" framing):

- Temporal misalignment between modalities (e.g. news arrives asynchronously relative to price ticks)
- Modality imbalance — one signal dominates training while others are effectively ignored
- Noisy or missing data
- Poor cross-market generalization

**Trends flagged for future work:** adaptive fusion, learning under missing modalities, integration of large language models, and temporal graph neural networks.

**Why it matters for your project:** Your current pipeline is strictly **unimodal** — OHLCV, VWAP, India VIX, and technical indicators only, with no news or sentiment data. Use this survey to consciously scope your paper: state explicitly that you are unimodal by design for this work, cite the survey as evidence that multimodal fusion is the field's current frontier, and mark it as future work. This preempts a reviewer asking why sentiment or news data was not incorporated.

# Part 2 — Research Roadmap

Current status: the full pipeline (8 stages, 13 trained models including Optimized Moirai) runs end-to-end and passes all leakage checks, but on the currently *synthetic* dataset, direction accuracy for every single model — including Optimized Moirai — clusters at or below the naive baseline (~0.28-0.32, close to chance for a 3-way up/flat/down split). The manuscript itself (figures, tables, write-up) has not yet been started. This is not a dead end; it is the normal starting point for this kind of project.

## Step 1 — Get the real dataset and re-run the full ladder

Swap `data/raw/raw_data.csv` for the faculty-supplied real NSE/market data. The pipeline needs no code changes if column names match ( `timestamp, open, high, low, close, volume, ...` ). Re-run all 8 stages and regenerate `results/tables/full_model_ladder_results.csv` . Every later step depends on this being real market data, not synthetic.

## Step 2 — Diagnose why direction accuracy sits at chance level

Before touching the model architecture, check: the class balance of `y_direction` (close to 33/33/33, or skewed?), the value the `fit_direction_threshold()` function picks for tau, and whether a 45-minute context window is simply too short to carry signal for a 60-minute-ahead prediction. This diagnosis is not a detour — it is a legitimate section of the eventual paper. “Why naive direction forecasting is hard at this horizon” is a real, citable finding.

## Step 3 — Reposition Optimized Moirai using Moirai-MoE’s argument

Explicitly frame the regime-embedding addition as a lightweight, interpretable alternative to full sparse-MoE routing, appropriate for a single-market, single-frequency setting. Cite Moirai-MoE directly for this contrast. This turns the architectural choice from something that looks arbitrary into something that is deliberately justified against the literature.

## Step 4 — Add a second evaluation target beyond raw direction accuracy

Given that results are likely to stay near chance on strict 3-way direction, also report the probabilistic metrics already computed (CRPS, 80% coverage), and consider a return-magnitude-weighted accuracy or a simplified 2-class (up/down, dropping “flat”) formulation. A model that is uncalibrated on direction but well-calibrated on its quantile forecasts is still a publishable, honest result.

## Step 5 — Run statistical significance tests across models

With test sets of only a few hundred rows, a few percentage points of accuracy difference between models may not be statistically meaningful. Add a paired test — McNemar’s test for direction accuracy, or a Diebold-Mariano test for forecast error — so the model comparison table reports p-values, not just raw numbers. Reviewers will ask for this.

## Step 6 — Write the related-work section now, before further coding

Enough material already exists to draft this: positioning against Moirai 1.0 and 2.0’s architectural choices, against Moirai-MoE’s specialization philosophy, and against the multimodal survey’s scope. Drafting this early forces clarity on what the actual contribution is, before more time is spent on code that may not end up supporting the final narrative.

## Step 7 — Pick a target venue and format early

Decide whether this is a conference paper (shorter, needs one crisp contribution), a journal submission (*Information Fusion*, the venue of the survey paper above, is a plausible fit given the topic), or a student/workshop paper. This decision changes how much ablation, breadth, and polish is needed before submission.

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**Note:** Do not treat “beat every baseline” as the immediate goal. With the current synthetic data, near-chance accuracy across the *entire* ladder — not just the proposed model — suggests Step 2 (diagnosis) matters more right now than any architecture change. Resolve that with real data first; the Moirai-MoE-style repositioning in Step 3 then becomes the paper’s actual contribution, rather than a footnote.